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Title: General Principles of Vaccination in Immunocompromised Paediatric Patients

Objective:

To provide clinical guidance on the use of live vaccines in immunocompromised patients

Policy Statement:

- This document provide general guidance on the use of vaccinations in immunocompromised pediatric patients

Policy Scope:

- Doctors and nurses dealing with vaccination in immunocompromised patients

Outlines:

Live vaccines in:

- Solid organ transplant
- Bone marrow transplant
- Post immune-ablative therapy
- Post steroids treatment
- Measles vaccine in HIV patients
- Vaccinating sibling of immunocompromised patients
- Management of exposure to measles
- Interval between immunoglobulin and live vaccines
- Mumps and Rubella vaccines and immunocompromised patients
- Management of varicella exposure in immunocompromised patients

Live vaccine in solid organ transplant recipient:

- Current recommendation is that measles vaccine can be administered as of 1 year post-transplant if no immunosuppression

Live vaccine for patients with bone marrow transplant:

- Bone marrow transplant can receive measles vaccine after 1.5 to 2 years post transplantation if they do not have chronic graft versus host disease

Live vaccine for patients with malignancy:

- Measles vaccine may be safely administered to patients with malignancies if they are in remission and have being off chemotherapy for at least 3 to 6 months
- Patients with Hodgkin's lymphoma have varying degree of long term T-cell deficit : Vaccination with MMR is not recommended for these patients

When a live vaccine can be given post immunoablative therapy?

- From 6 to 12 months (APP Red book: recommend 3 months but there is no firm recommendations)

When an inactivated or component vaccines can be given post immunoablative therapy?

- can be given as soon as total lymphocyte count exceeds 500
- can be given during maintenance chemotherapy
- Booster required > 3 months after the end of therapy

What is high dose steroids?

- High dose steroids is more than or equal to 2 mg per KG/day of prednisone or equivalent to a total of more than or equal to 20 mg /day for children weighing more than 10 KG

When a live vaccine can be given post steroids therapy?

- 1- for high dose steroid for more than 2 weeks: duration of delay four weeks
- 2- for high dose steroid for less than two weeks: delay of delay 2 weeks
- There should be no delay in administering live vaccines of the patient getting: topical (skin or respiratory tract) , local injection, physiologic dose or low or moderate systemic dose

Why an HIV infected patients can receive measles vaccine but not some categories of transplant recipient?

- Data specific to the HIV infected group indicate that the risk of the children dying from measles is far greater from vaccine –related infection
- They can receive MMR vaccine if the CD4 count > 200 for at least more than 6 months

IF a sibling of transplant /immunocompromised children vaccinated with measles, is there any chance of measles virus being transmitted as a result of vaccination from these children ?

- No, measles vaccine will not result in transmission of vaccine related measles virus. It is important to vaccinate siblings against measles as this will reduce the risk that the transplant recipient will become expose to wild measles vaccine at home

What should be done if a transplant recipient or immunocompromised patient is exposed to measles?

- First varify that the **index** case is in fact confirmed measles case
- For exposed patients : IM immune globulin 0.5 ml/kg (max 15 ML) regardless of their vaccination status

What is the interval between immune globulin use and measles vaccine:

- It differs between **different** indications for the immune globulin : refer to the APP red book)

How soon after measles (live vaccine) vaccination can children undergoes solid organ transplantation/receive immunosuppressive medications?

- Usually after 4-6 weeks following vaccinations

How soon after non-live vaccine can children undergoes solid organ transplantation/receive immunosuppressive medications?

- At least two weeks

What is earliest age at which the measles vaccine can be given pre-transplant to solid organ transplant candidate?

- The current recommendation is that the vaccine can be given as early as 6 months of age

Can mumps vaccine virus be transmitted?

- Vaccinated persons do not transmit mumps vaccine virus

Can Rubella vaccine virus be transmitted?

- Most vaccinated persons intermittently shed small amount of rubella virus from pharynx 7-28 days after vaccination. However there is no evidence of transmission of the vaccine virus in studies of susceptible household contact

Who is the high risk patient to consider varicella immunoglobulin as post exposure prophylaxis?

- - Patients with evidence of severe primary immunodeficiency, for example, severe combined immunodeficiency (SCID), Wiskott-Aldrich syndrome and other combined immunodeficiency syndromes
- -All patients currently being treated for malignant disease with immunosuppressive chemotherapy or radiotherapy, and for at least **six months** after terminating such treatment
- -all patients who have received a solid organ transplant and are currently on immunosuppressive treatment
- -patients who have received a bone marrow transplant until at least 12 months after finishing all immunosuppressive treatment, or longer where the patient has developed graft-versus-host disease, (18 months for non-sibling, unrelated or mismatched donors)
- -all patients receiving systemic high-dose steroids until at least three months after treatment has stopped.
- -patients receiving other types of immunosuppressive drugs (e.g. azathioprine, ciclosporin, methotrexate, cyclophosphamide, leflunomide and the newer cytokine inhibitors) alone or in combination with lower
- -patients with immunosuppression due to HIV infection WITH CD4 count less than 15%

What are the benefits of giving varicella immunoglobulin after exposure for high - risk patients?

- VZIG has been shown to modify varicella infections.
- VZIG may prevent infection altogether or result in milder disease.
- The effect of VZIG is maximized by giving it within 72 hours of exposure but can be given up to 10 days post contact

References:

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Vaccination of
immune
compromised
children—an
overview for
physicians

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